

RAJAN KUMAR JHA

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Summary

High-Performance Backend Specialist with 3+ years of experience in architecting scalable systems using Python, Django, and Golang. Proven track record of reducing API latency by 99.9% (30s to 100ms) and building complex financial reconciliation engines. Expert in Database Optimization (Redis, ClickHouse), GenAI integrations, and high-concurrency system design. Passionate about solving complex engineering problems and optimizing backend infrastructure.

Skills

Languages: Python (Expert), Golang, SQL, C++

Backend: Django, FastAPI, Django REST Framework, REST APIs, WebSockets, Celery, Apache Kafka

Databases: PostgreSQL, MySQL, Redis, ClickHouse, MongoDB, MSSQL

AI & NLP: SpaCy, FinBERT, DistilBERT, Amazon Bedrock, RAG, NLP-to-SQL

Cloud & DevOps: AWS (EC2, RDS, Bedrock), Docker, Kubernetes, Helm, Nginx, Azure, Git, GitHub Actions, CI/CD

Architecture System Design: Microservices, Event-Driven Architecture, Circuit Breaker, Caching, Rate Limiting, Async Processing, Query Optimization, Distributed Systems, API Performance, Security

Testing: Pytest, Unit Testing, Integration Testing

Experience

OpsTree Solutions

Backend Developer Specialist

Noida, India

Aug 2024 – Present

Client: LUXASIA

- Solely architected and implemented a custom finance reconciliation platform using Django and MSSQL, streamlining the ingestion of financial reports across Shopee, Lazada, TikTok, and Amazon.
- Built dynamic file upload workflows via customized Django Admin and Django Forms, enabling platform-specific validations and marketing fee entry.
- Engineered data transformation pipelines using Pandas and openpyxl to parse, clean, and normalize Excel data, supporting automated reconciliation.
- Automated complex currency conversion and tax extraction logic (VAT, GST) for country-wise uploads, significantly improving financial data accuracy.
- Reduced manual processing time from 20 days (by 4 analysts) to near-instant, enabling faster insights and operational efficiency.
- Generated unified reports combining multi-platform data using backend logic and saved them with version control for audit trails.
- Ensured data validation through reusable schema checks and custom validators, ensuring 100% data integrity.

Tech Stack: Python, Django, MSSQL, Pandas, OpenPyXL, Docker, Azure.

Client: Careers360

High-Performance Platform Optimization [Live Project]

- Critical Achievement: Reduced Signup/Login API latency from 60-90 seconds to 40-50 milliseconds (99% improvement) using database query optimizations and caching strategies.
- Integrated Google Social Signup and optimized the authentication flow, reducing onboarding friction.
- Implemented aggressive Redis caching and async response strategies to handle peak load.
- Enabled the platform to handle 35K+ concurrent users and 20K+ new registrations on JEE Main Day-1, resulting in a 700% rise in new user signups.

Rank Prediction Engine [Live Project]

- Built a predictive ranking engine from scratch using Django, MySQL, Redis, and ClickHouse to estimate user percentile for nationwide exams (JEE, NEET).
- Designed logic to compute multi-scenario outputs by caste and disability categories (GEN, OBC, PWD).
- Enabled bidirectional prediction flows: Score → Rank and Percentile → Rank, customized for each exam's scoring methodology.

- Scaled backend to handle real-time traffic, serving over 17,000+ users/hour on exam result days.

College Prediction System [Live Project]

- Designed the backend of a dynamic College Prediction System matching students to colleges based on rank, category, and preferences.
- Developed a logic engine to handle multi-reservation mapping and suggest accurate college options across exams like JEE Main, WBJEE, COMEDK.
- Built scalable DB schemas and optimized queries to support filter-based search (fees, sports, placements).
- Improved scalability using Redis-based caching, enabling smooth load handling.

Tech Stack: Python, Django, Redis, ClickHouse, MySQL, Celery, AWS.

Internal Tooling (DevOps & Go)

- Developed high-concurrency internal log analysis tools using Golang, enabling the DevOps team to parse and analyze server logs 5x faster than previous implementations.

Tech Stack: Golang, Cobra CLI, File I/O.

Captwist.in

Software Development Engineer

Remote
July 2022 – July 2024

Project: Financial News Platform

- Designed and implemented a real-time corporate news Feed system using Django and PostgreSQL, delivering categorized financial news across companies and sectors.
- Automated data ingestion pipelines from multiple financial sources.
- Applied FinBERT + SpaCy for sentiment classification and entity recognition to tag news items automatically.
- Built robust backend APIs to serve news feeds filtered by user preferences, news type, and sectors, improving engagement and relevance.
- Applied ML-based logic for content personalization and relevance scoring, leading to higher user retention.
- Optimized database access patterns and caching layers to handle high ingestion rates.

Tech Stack: Django, PostgreSQL, FinBERT, SpaCy, AWS EC2, Redis.

Project: User Management System

- Developed secure and efficient user authentication system using OAuth 2.0, JWT, and 2FA, improving platform security.
- Enabled social logins (Facebook) to enhance signup flow and reduce onboarding friction.
- Built a real-time notification system to deliver personalized alerts and activity updates.
- Integrated Razorpay for in-app transactions and managed secure payment flow.
- Improved backend API performance by profiling Django ORM and optimizing query execution.

Tech Stack: Django Rest Framework, OAuth2, JWT, Razorpay SDK, Firebase.

Technical Projects

AI-Powered Chatbot & SQL Intelligence System | Python, FastAPI, Bedrock, OpenSearch

- Built an end-to-end intelligent system combining LLM-driven education chatbot and NLP-to-SQL query translator to assist non-technical users in accessing structured insights from unstructured and tabular data.
- Developed a domain-specific chatbot using Bedrock LLM, FastAPI, and Streamlit, capable of answering queries about colleges, courses, placements, etc., based on scraped and processed educational data.
- Designed a scalable knowledge base architecture using OpenSearch, and enabled contextual response retrieval with Redis and WebSocket-based live UI.
- Used SpaCy and DistilBERT for text preprocessing, sentiment tagging, and summarization; leveraged Boto3 to integrate Amazon Bedrock with streaming Gradual Reveal output.
- Integrated an NLP-to-SQL engine using SpaCy for entity and condition extraction, and mapped parsed text to SQL clauses (SELECT, WHERE, ORDER BY) via Python logic.
- Enabled natural language access to databases - allowing users to run complex SQL queries using plain English input.
- Ensured modularity and extensibility across components using Pydantic validation, clean code principles, and multi-dialect SQL compatibility.

Education

Maulana Abul Kalam Azad University of Technology

B.Tech in Information Technology (CGPA: 8.52)

West Bengal

2019 – 2023

Oriental College Patna City

Senior Secondary (PCM) - 76.8%

Patna, Bihar

2019